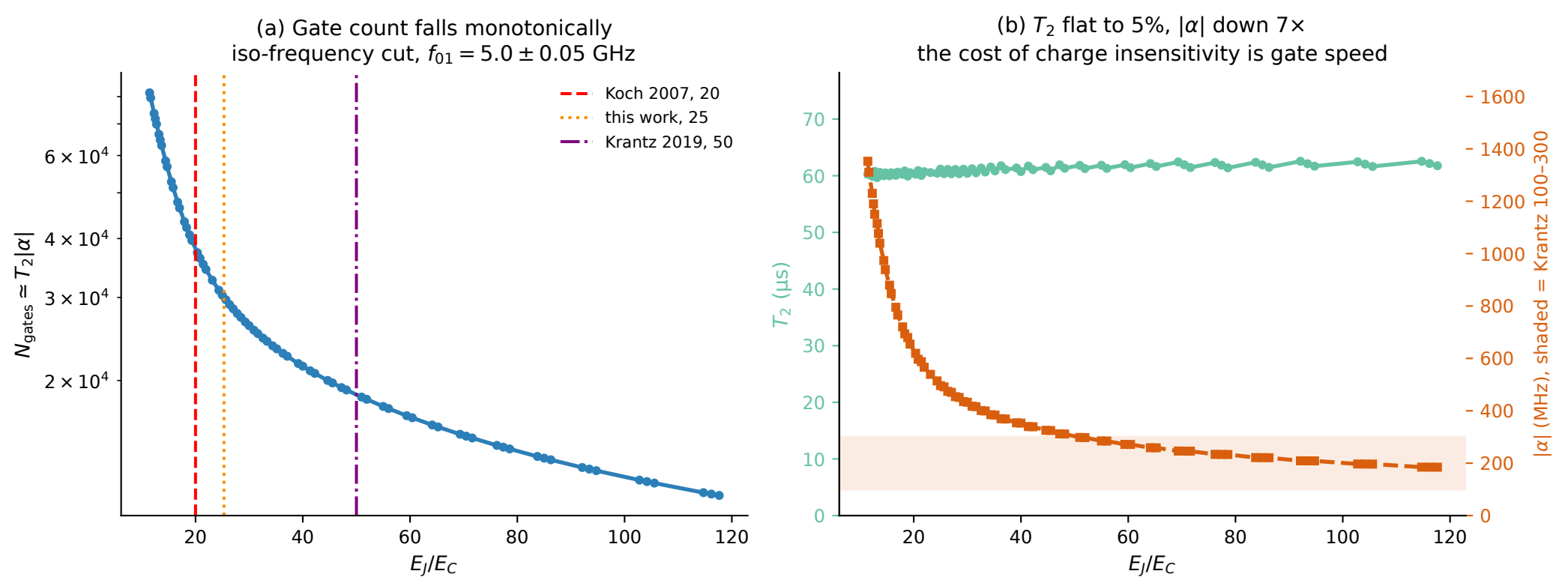

(a) Gate count falls monotonically
iso-frequency cut, $f_{01} = 5.0 \pm 0.05$ GHz
Koch 2007, 20
this work, 25
Krantz 2019, 50
$N_{gates} \simeq T_2|\alpha|$
6×10^4
4×10^4
3×10^4
2×10^4
20
40
60
80
100
120
E_J/E_C
(b) T_2 flat to 5%, $|\alpha|$ down 7×
the cost of charge insensitivity is gate speed
T_2 (μs)
0
10
20
30
40
50
60
70
$|\alpha|$ (MHz), shaded = Krantz 100–300
0
200
400
600
800
1000
1200
1400
1600
20
40
60
80
100
120
E_J/E_C